What are we trying to forecast?

A guided revision of the task, networks, experiments and evidence

Our question

From the tactical observer's available history, which catalogue label will a trained pilot select at +1, +2 or +5 seconds?

Understand

Task, data and inputs

Explain

Networks and training

Assess

Results and uncertainty

Main finding: the current MLP/TCN reference retains higher measured accuracy than the pure Transformer challenger.

CHECK What information goes into the forecaster, and what exactly comes out?

Two different systems learn two different jobs

The RL pilot acts; the supervised forecaster predicts its choices

1 Train the pilots with RL

Simulator → observations → pilot actions
Reward teaches the policy how to act.
Freeze ten trained policies.

2 Train a skill forecaster

B01's observed tracks → forecast labels
Supervised examples teach prediction.
One shared forecaster covers all ten pilots.

What a forecast means

A01 is the observed aircraft; B01 is the scripted tactical observer/opponent. A predicted label describes the pilot's catalogue choice. Forecast accuracy does not rate combat competence.

CHECK How does training the RL pilot differ from training its skill forecaster?

Why 3,000 encounters after 112,500 training flights?

Pilot training and forecasting data collection are separate stages

Pilot training

112,500 simulated training flights
A population-training stage develops the policies.
These are not 112,500 forecasting examples.

Forecasting corpus

300 scenario groups × 10 frozen pilots
= 3,000 observer–pilot encounters
Each encounter supplies overlapping origins.

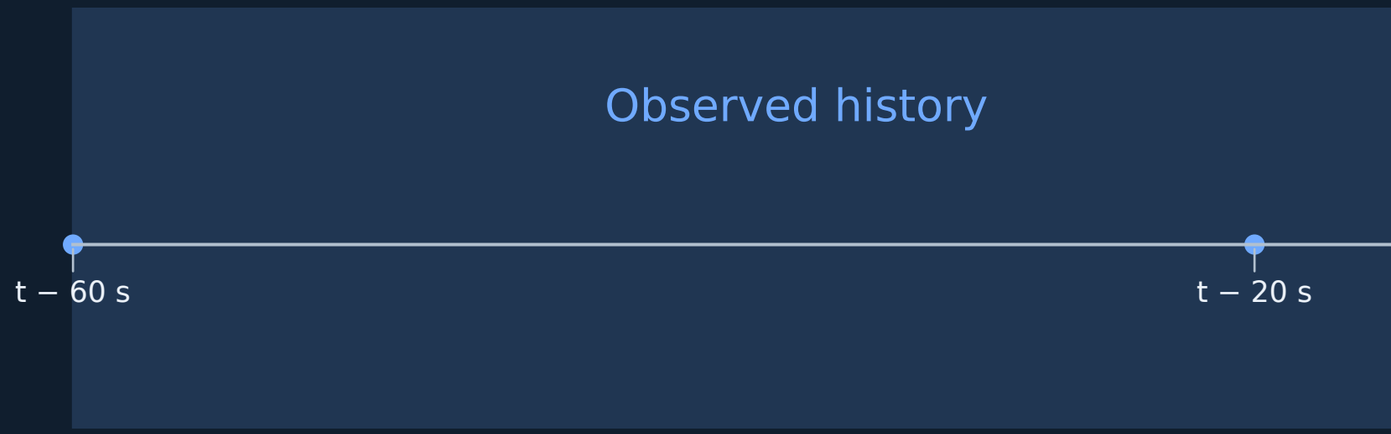
Partition	Scenario groups	Encounters	Purpose
Training	210	2,100	Fit and internal model selection
Validation	45	450	Fixed development comparisons
Reserved test	45	450	Not evaluated

Keep all ten versions of a scenario together. Thousands of overlapping windows are not thousands of independent scenarios.

CHECK Why are the RL training-flight count and the forecasting encounter count different?

History, forecast horizon and elapsed time

Three clocks that must stay distinct



History length: how far back the model may look.

Forecast horizon: how far ahead the label lies.

Elapsed time: how old the encounter is at this origin.

Early origins remain in the evaluation. Missing earlier history is masked; attention never receives future tracks or future labels.

CHECK Does using 60 seconds of observation history introduce future information?

What observations does the forecaster actually use?

54 original features plus 37 derived geometric features

Feature group	Examples used in forecasting
Observer and relative tracks	B01 position/velocity/heading; A01 relative position/velocity, range and bearing
Missiles and energy	Hostile-missile flag; four B01 missile slots; kinetic/potential-energy proxies
Backward changes	Target acceleration derived from observations 0.5 seconds apart
37 geometry features	Line-of-sight rates, aspect, closure/closest approach, relative energy, missile presence

54 original + 37 derived = 91 features

Derived geometry uses available observations. Pilot ID and the private current skill label are not inputs.

CHECK Give three concrete forecasting features, and one item that is deliberately unavailable.

A 3D history can become an ordinary feature row

Batch × observation time × feature

Time	Speed	Range	Valid
t − 2	250	24,000	1
t − 1	255	23,600	1
t	260	23,200	1



One ordinary feature row

[speed(t−2), range(t−2), valid(t−2), …,
speed(t), range(t), valid(t)]
Time keeps a fixed column position.

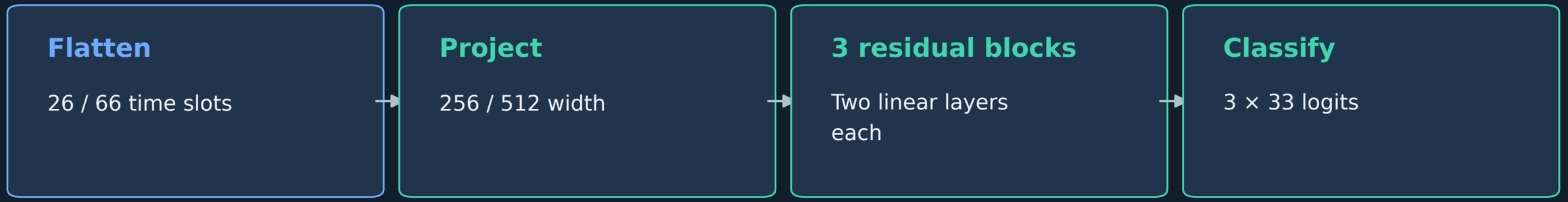
Batch × time × features is a stack of those per-origin tables.
20-second MLP: 26 timestamps × (91 features + mask) = 2,392 columns.
60-second MLP: 66 × 92 = 6,072 columns.

Toy values above are illustrative. The target is a separate three-column table of future catalogue labels.

CHECK How does a temporal MLP turn a history tensor into a 2D training table?

The temporal MLP and the twelve-model ensemble

Ordered history → dense computation → three label distributions



Eight linear layers overall: input projection + six inside the residual blocks + output projection. A residual connection adds a block's input back to its output.

The v3 ensemble

Four history/width/regularization recipes × three random training seeds = twelve members.
Their calibrated probabilities are averaged equally.

CHECK What makes this MLP temporal, and why are there twelve models?

The TCN adds a different view of temporal patterns

The v4 incumbent combines MLP and convolutional predictions

Temporal convolutional network

Six causal, dilated convolution blocks
Dilations: 1, 2, 4, 8, 16, 32
127-token receptive field covers all 66 input tokens.

Current reference: 15 members

12 temporal MLPs → mean probabilities
3 sixty-second TCNs → mean probabilities
Mix those family means 50 : 50.

P(final label) = 0.5 × mean(P from 12 MLPs) + 0.5 × mean(P from 3 TCNs)

Illustrative family predictions	Label A	Label B	Label C
MLPs / TCNs	0.60 / 0.20	0.30 / 0.60	0.10 / 0.20
50 : 50 mixture	0.40	0.45	0.15

CHECK Is v4 a uniform average of fifteen model outputs?

What the six new Transformers do

Three temporal Transformers plus three patch Transformers

Temporal Transformer

66 observed tokens → attention
Retain all contextual positions in readout
Three training seeds = three members

Patch Transformer

66 observed tokens → 17 local patches
Attention combines the patch representations
Three training seeds = three members

Each encoder: 3 attention blocks • width 192 • 4 attention heads
Current observed features also enter the classifier. Each member outputs 3 × 33 logits.
The final six-member ensemble averages calibrated probabilities equally.

Attention connects observations inside the available past. Standard feed-forward sublayers are part of a Transformer.
These models are trained from scratch.

CHECK How do the temporal and patch Transformers differ?

Four milestones explain the experiment history

Keep the comparison focused on the main systems

Main milestone	Features	History	Input tokens	Members	Parameters
Original Transformer	54	5 s	51	1	0.44 M
12 MLPs (v3)	91	20/60 s	26/66	12	28.98 M
MLP + TCN (v4)	91	20/60 s	26/66	15	31.73 M
6 Transformers (v5)	91	60 s	66	6	18.91 M

Original: 51 samples in five seconds. Later models: sparse sampling over 20/60 seconds. Longer history does not always mean more input tokens.

Several design choices changed together. The history, features, architecture and model count are explicit; these are complete-recipe comparisons.

CHECK Why does the experiment history not isolate a single winning architectural ingredient?

Which data can influence each training decision?

Group scenarios first; separate fitting, stopping and recipe selection

147 fit groups

Weights and normalization

21 stop groups

Epochs and calibration

42 select groups

Compare candidate recipes

These groups all sit inside the original 210 training groups. Repeat across two internal partitions; each has all ten pilots for every included scenario.

Freeze → refit on all 210 → evaluate on the same 45 validation groups

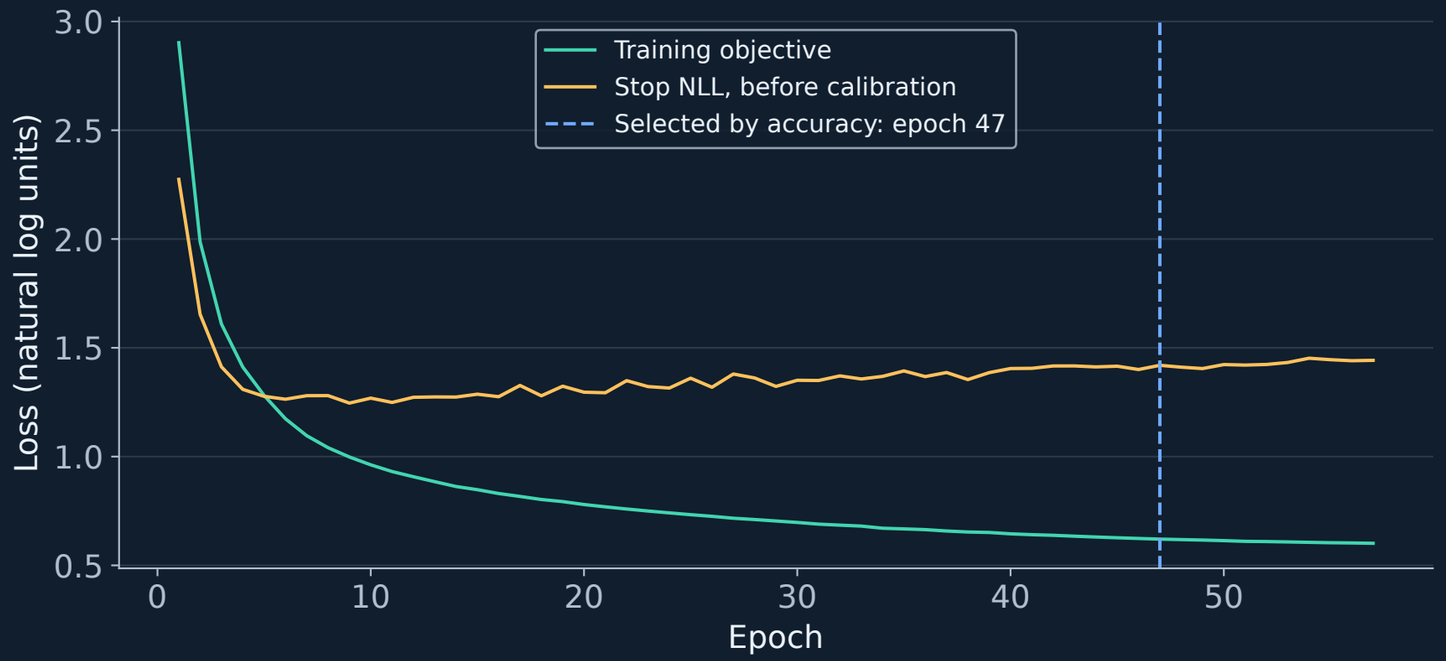
Freeze members, seeds, epoch budgets and temperatures first. Keep the 45-group reserved test separate.

The original neural models used validation for stopping; validation has been inspected repeatedly. Current scores are development evidence.

CHECK What are the different jobs of fit, stop, selection, validation and reserved-test groups?

Cross-entropy, NLL and the learning curves

Accuracy can improve while confident mistakes become more costly



NLL = −ln(p of the true label)

p(true label)	NLL
0.90	0.105
0.50	0.693
0.10	2.303

Lower is better. A confident mistake is expensive. NLL is not a percentage.

Actual 60 s patch model, internal partition 0, seed 42. Training uses label smoothing and dropout; stopping NLL uses ordinary evaluation predictions. The loss gap is not directly comparable.

Forecaster curves pool pilots. Per-pilot forecaster epoch curves were not logged; the archived RL-return curves describe a different learner.

CHECK What does NLL measure, and why can the plotted losses disagree?

Accuracy, precision, recall and F1

One prediction question, several complementary measurements

Across all predictions

Top-1: was the highest-ranked label correct?
Top-3: was the true label among three suggestions?

For one catalogue label

Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 = 2 × precision × recall / (precision + recall)

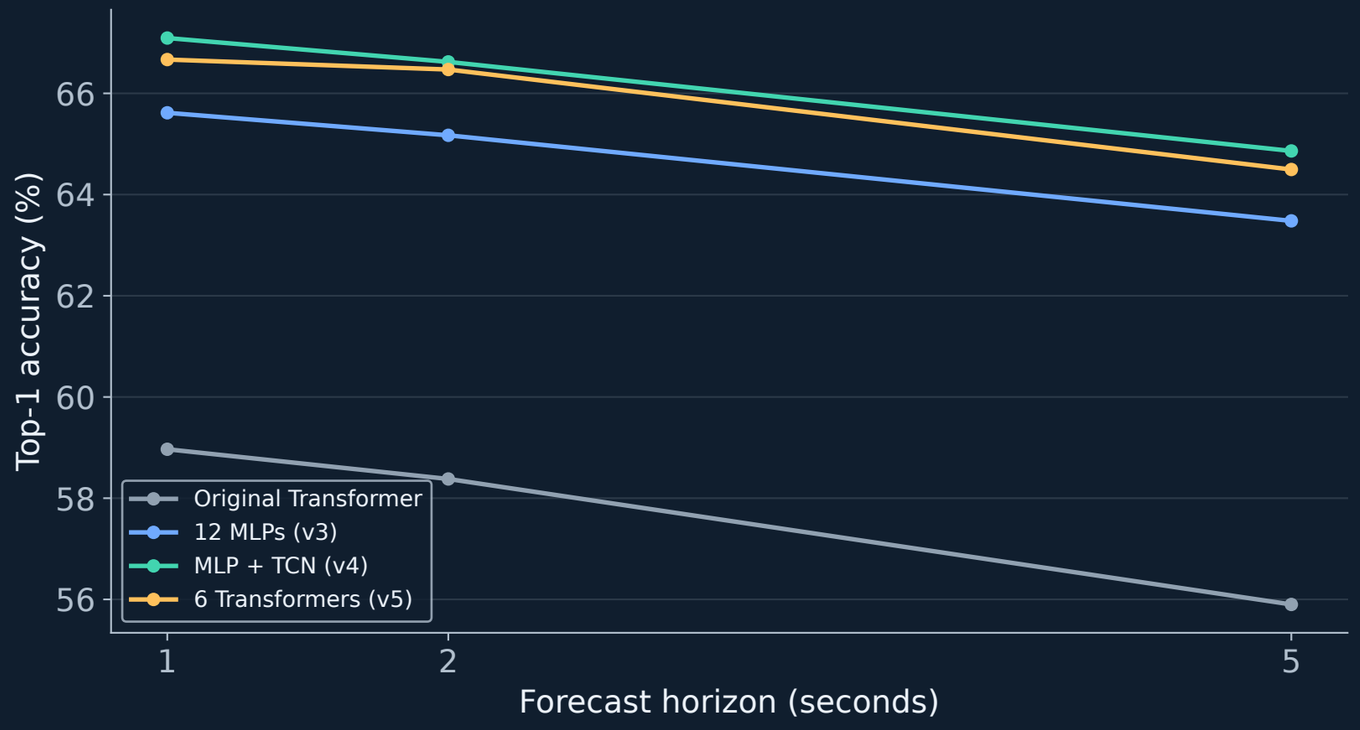
Illustrative counts	Precision	Recall	F1
60 true positives; 20 false positives; 40 missed	60 / 80 = 75%	60 / 100 = 60%	66.7%

Catalogue macro-F1 averages the 33 class F1 values equally, including zero-F1 absent labels. It does not weight frequent labels more heavily.

CHECK With TP=60, FP=20 and FN=40 for one skill, what are precision, recall and F1?

How the main models compare on accuracy

Same forecast origins, labels and +1/+2/+5-second targets



Model	+1 s	+2 s	+5 s	Mean
Original Transformer	58.97%	58.38%	55.90%	57.75%
12 MLPs (v3)	65.62%	65.17%	63.48%	64.76%
MLP + TCN (v4)	67.09%	66.62%	64.86%	66.19%
6 Transformers (v5)	66.67%	66.47%	64.50%	65.88%

Same origins, 33 labels and target masks.
The mean weights the three horizons equally.

Validation support: 30,802 / 30,466 / 29,458 eligible origins at +1 / +2 / +5 s. Missing future labels are masked identically.

CHECK What is the main accuracy finding from the Transformer experiment?

Accuracy is only one part of the scorecard

+5-second probability quality, class balance and model size

Model, +5 s	Top-3 ↑	F1 / 33 ↑	NLL ↓	Brier ↓	ECE ↓
Original Transformer	86.51%	0.2739	1.2903	0.5830	8.74 pp
12 MLPs (v3)	89.25%	0.3665	1.1642	0.4916	3.42 pp
MLP + TCN (v4)	89.86%	0.3767	1.1243	0.4763	4.36 pp
6 Transformers (v5)	90.14%	0.3639	1.0980	0.4744	3.82 pp

What v5 improves

NLL, Brier and ECE improve versus v4. It uses 18.9 M rather than 31.7 M parameters.

What v5 loses

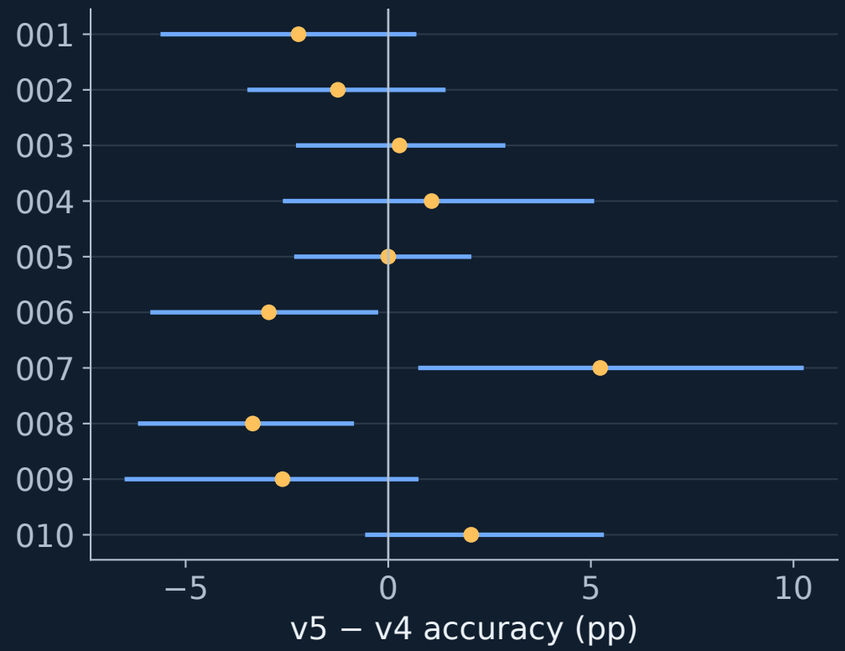
Top-1 and macro-F1 fall at all three horizons. Better probabilities need not produce more correct top labels.

CHECK How can the Transformer be useful even when its top-one accuracy is slightly lower?

Which trained pilots are easier to forecast?

+5-second accuracy and changes for all ten pilots

Pilot	Origins	v4	v5	Gain pp
001	2,845	55.85%	53.64%	-2.21
002	3,057	76.94%	75.70%	-1.24
003	2,878	65.46%	65.74%	+0.28
004	2,897	69.38%	70.45%	+1.07
005	3,049	74.52%	74.52%	+0.00
006	2,987	65.38%	62.44%	-2.95
007	3,001	54.88%	60.11%	+5.23
008	2,871	45.35%	42.01%	-3.34
009	2,990	71.77%	69.16%	-2.61
010	2,883	67.71%	69.75%	+2.05



Four pilot point estimates improve at +5 s. The 95% paired scenario intervals are conditional and unadjusted for multiple comparisons. Accuracy is not pilot competence.

CHECK Does a higher forecast accuracy mean the underlying pilot is more competent?

Per-skill F1 needs its support count

+5 seconds: all 33 catalogue labels remain in the comparison

Skill	Support	v4 F1	v5 F1
abort_support	113	0.515	0.506
accelerate_to_speed	0	—	—
beam_missile_left	6	0.000	0.000
beam_missile_right	20	0.513	0.513
beam_target_left	1,693	0.725	0.721
beam_target_right	1,445	0.737	0.724
climb_to_altitude	1	0.000	0.000
commit	0	—	—
crank_target_left	1,150	0.653	0.646
crank_target_right	947	0.788	0.776
decelerate_to_speed	126	0.307	0.284
defensive_reversal	1	0.000	0.000
descend_to_altitude	162	0.631	0.649
dive_defense	0	—	—
drag_missile	0	—	—
extend	2,676	0.602	0.584
lag_pursuit	4,872	0.632	0.638

Skill	Support	v4 F1	v5 F1
last_ditch_break	0	—	—
launch	130	0.507	0.497
lead_pursuit	1,350	0.543	0.533
lock_target	0	—	—
maintain_heading	36	0.500	0.575
notch_left	124	0.430	0.420
notch_right	80	0.462	0.439
preemptive_crank	0	—	—
pursue_target	2,561	0.562	0.561
recommit	4,857	0.666	0.656
search	2	0.000	0.000
secondary_shot	1,414	0.656	0.651
short_range_attack	216	0.291	0.268
support_missile	19	0.429	0.160
turn_cold	5,437	0.701	0.706
turn_to_heading	20	0.585	0.500

Dash = no true examples at +5 s. Those labels still count as zero in the all-33 macro-F1. Support counts overlapping origins, not independent flights.

Example: support_missile has only 19 origins; its F1 falls from 0.429 to 0.160. Interpret sparse-label changes cautiously.

CHECK Why must a skill's F1 score be read beside its number of true examples?

What the temporal statistics actually tell us

Forecast difficulty, repeated labels and changing encounter populations

Elapsed phase	Encounters	Origins	v4 +5 s	v5 +5 s
6-20 s	450	6,411	58.13%	57.87%
21-60 s	389	11,531	65.85%	65.72%
61+ s	225	11,516	67.62%	66.96%

Selected labels often persist

91.77% of adjacent observed decisions repeat the same label. Completed runs: median 5 s, 90th percentile 16 s.

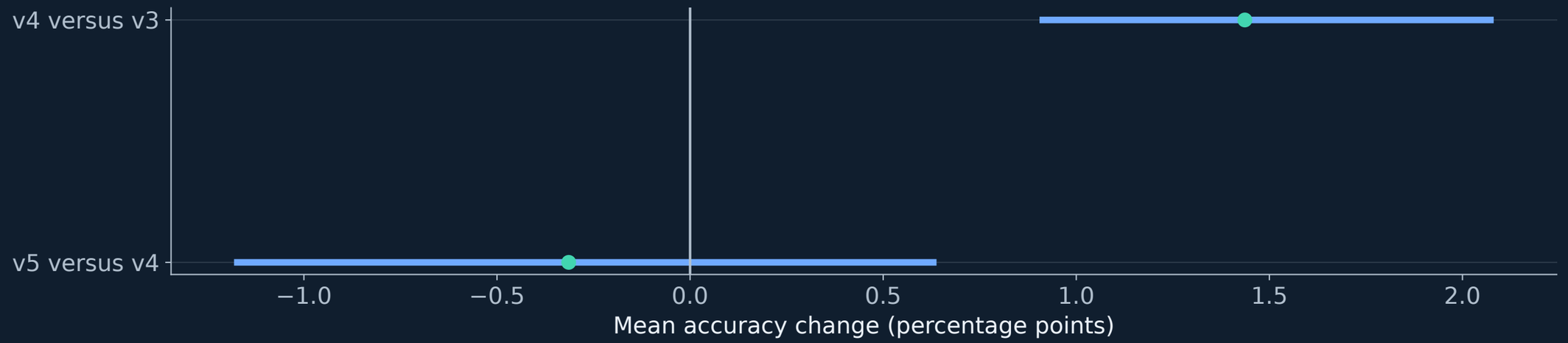
Common-origin analysis requires a valid +5 s target for every compared horizon. Later phases contain a different mix of surviving encounters and decisions.

Later accuracy does not isolate the value of extra history. Label persistence is descriptive: the private current true label remains unavailable to the forecaster.

CHECK Why is the observed 91.77% one-second repeat fraction not a deployable forecasting baseline?

How certain are the measured differences?

Resample whole shared scenarios, not overlapping windows



95% intervals from 5,000 paired resamples of whole scenario groups preserve the ten pilots and correlated horizons. They are conditional development intervals, not final-test evidence.

Secondary frozen blend: 75% v5 + 25% v4, 21 models. Mean gain +0.07 pp; interval [-0.63, +0.84]. It establishes no accuracy gain or equivalence.

CHECK What does it mean when the 95% interval for an accuracy difference crosses zero?

How much experiment work has been done?

Count distinct fits and state what their timers include

Study	New completed fits	Summed fit timers	Internal	Final refits
12 MLPs (v3)	23	24.31 min	11	12
MLP + TCN (v4)	37	57.09 min	34	3
6 Transformers (v5)	29	135.30 min	23	6

A fit is one recipe × one seed × one partition (or a final refit). Cached reference models are reused and excluded from these new-fit counts.

The v3 row covers v3 alone; it excludes the original two neural fits and eleven earlier v2 fits. v4/v5 timers include shared-GPU contention and exclude loading/setup.

Summed fit minutes are not elapsed project time, simulation cost or an isolated speed comparison between architectures.

CHECK Why should the summed fit minutes not be read as total elapsed project time?

What remains uncertain, and what could improve?

Keep future hypotheses separate from completed evidence

Evidence boundaries

Known pilots under new scenarios
Idealised simulator tracks
Selected catalogue labels, with some aliases
Rare/absent labels and uneven pilot effects
Repeatedly inspected development validation

Prospective experiments

Stronger regularization and probability-aware stopping
Training-only representation learning
Controlled feature/history ablations
Performance-versus-cost objectives
Final test after choices settle

Freeze a new experiment before inspecting its validation results. Keep the existing task and baseline comparable, and report failed candidates and subgroup regressions.

These are future hypotheses. The current revision deck reads saved evidence only; it performs no new training, inference or test evaluation.

CHECK What would make the next performance claim more convincing?

Revision recap: what can we now explain?

Task → data → network → training → evidence

What the experiments establish

The task predicts future skill selections from observed history. MLP/TCN models can use temporal context effectively. The new pure Transformers are smaller and improve probability quality; they do not establish improved top-1 accuracy.

Revision step	What you should be able to say
Explain the task	Observer → history → three future labels
Explain the model	Input shape → temporal computation → probabilities
Judge the evidence	Accuracy + F1 + NLL + pilot/skill support + uncertainty

CHECK Can you explain the complete experiment in your own words, from B01's history to the reported result?